

Social Media Misinformation and Voting Intentions: Older Adults' Experiences with Manipulative Narratives

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Older adults habitually encounter misinformation, yet little is known about their experiences with it. In this study, we employed a mixed-methods approach, combining a survey ($n = 119$) with semi-structured interviews ($n = 21$), to investigate how older adults in America conceptualize, discern, and contextualize social media misinformation. Given the historical context of misinformation being used to influence voting outcomes, our study specifically examined this phenomenon from a voting intention perspective. Our findings reveal that 62% of participants intending to vote Democrat perceived a manipulative political purpose behind the spread of misinformation, whereas only 5% of those intending to vote Republican believed that misinformation serves a political dissent purpose. Regardless of voting intentions, most participants relied on source heuristics and fact-checking to discern truth from misinformation on social media. A major concern among participants was the biased reasoning influenced by personal values and emotions affected by misinformation. Notably, 74% of participants intending to vote Democrat were concerned that misinformation would escalate extremism in the future. In contrast, those intending to vote Republican, those undecided, or those planning to abstain expressed concerns that misinformation would further erode trust in democratic institutions, particularly in public health and free and fair elections. During our interviews, we discovered that 63% of participants intending to vote Republican mentioned that Republican or conservative voices often disseminate misinformation, even though these participants were closely aligned with this political ideology.

CCS Concepts: • **Security and privacy** → **Social aspects of security and privacy**; **Usability in security and privacy**; • **Human-centered computing** → **Social networks**; • **Social and professional topics** → **Seniors**.

Additional Key Words and Phrases: misinformation, older adults, truth discernment, social media, voting

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1 Introduction

Older adults, aged 65 and up [25], are increasingly active participants in digital interactions, with over 61% owning a smartphone and about half using social media [23]. Despite their growing engagement, prior studies indicate that this age group was twice as likely to be exposed to misinformation on social media during the 2016 US elections compared to younger counterparts [30]. Research has extensively explored the intersection of older adults, technology adoption, and digital

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literacy, particularly focusing on their approaches to handling security and privacy-related information online [49, 56, 57]. Older adults are generally proactive in protecting themselves from digital hazards, though they often lack sufficient support [55] and may rely on collaborative efforts to maintain their agency [49, 56]. While they may avoid social media for information security or privacy concerns, older adults actively use it for various other information-seeking purposes [57]. Their wealth of life experience and analytical skills significantly contribute to their digital interactions, particularly during voting scenarios, which can be severely impacted by misinformation.

Misinformation, lacking truth and truthfulness [39], encompasses fabrications, rumors, conspiracies, hoaxes, and clickbait [24]. This type of information is prevalent on social media, covering topics from climate change to health concerns and elections [48]. Given their interest in these topics, older adults are frequently exposed to misinformation. Moreover, the unpredictable consequences of social media misinformation on election results and public health have inspired crucial research at the intersection of older adults, social media adoption, and misinformation [5, 31, 52, 93]. Although older adults might initially be susceptible to fake news, their accumulated knowledge and analytical thinking skills help them evaluate claims over time. However, existing research primarily focuses on political news headlines in controlled settings, often overlooking how older adults conceptualize and interpret the intentionality and purpose of misinformation, particularly in the context of influencing voting intentions, and with that, affecting election outcomes.

To address this gap, we conducted a mixed-methods study combining a survey ($n=119$) with in-depth, semi-structured interviews ($n=21$) to investigate how older adults in America conceptualize, discern, and contextualize misinformation in relation to their voting intentions. Previous studies have focused on receptivity and truth discernment based on political beliefs [63, 75, 87]. In contrast, we chose to examine older adults' mental models of misinformation relative to their voting intentions for the forthcoming 2024 US presidential elections. This focus allows us to explore how older adults use misinformation to inform their voting decisions, avoiding potential stereotyping based on political ideologies amidst heightened political scrutiny of misinformation research [88].

Our results suggest that 63% of participants intending to vote Democrat in the 2024 US elections maintain a mental model of misinformation as information containing elements of truth for manipulative political purposes. Conversely, only 3% of participants intending to vote Republican see misinformation as primarily serving political dissent. Participants with varied voting intentions believe misinformation aims to create political divisions and manipulate emotions. A significant theme emerged from our data: skepticism about repairing the political damage caused by misinformation since the 2016 election cycle. Notably, five out of eight participants intending to vote Republican acknowledged that prominent conservative voices frequently disseminate misinformation, despite the fact that these voices closely align with their political ideology.

In summary, the **contributions** of our work are fourfold:

- (1) Firstly, we believe this is the first study to detail the mental models older adults employ when dealing with misinformation on social media. This novel insight enhances our understanding of their cognitive processes and strategies, contributing to the broader field of digital literacy and misinformation management among older adults as a vulnerable population at a heightened risk of being targeted with information operations online [80];
- (2) Secondly, we provide an in-depth account of the relevant factors older adults consider when encountering misinformation and forming their voting intentions. This includes the perceived origins and purpose of misinformation, cues of deception, reasoning strategies, and the broader social context of potential information operations. This analysis helps to inform the design of more effective interventions, educational tools, and supportive technologies that can assist older adults in navigating digital information online;

- (3) Thirdly, we highlight how older adults utilize a combination of source heuristics and fact-checking practices to discern truth from misinformation on social media. By understanding these methods, we can better tailor digital platforms and fact-checking resources to meet the specific needs of older adults, fostering a more inclusive digital environment that caters both to their needs but also involves them in the credibility assessment process of emerging topics often subject to manipulative interpretations;
- (4) Finally, we report on older adults' opinions regarding the impact of misinformation on American society to date and its potential future effects through a mixed-methods approach. This perspective is crucial for developing collaborative efforts and policy recommendations to address misinformation and its societal implications, particularly towards safeguarding free and fair elections. Additionally, our findings underscore the importance of addressing biased reasoning influenced by personal values and emotions and suggest strategies for mitigating the negative effects of misinformation on public trust and democratic processes.

2 Related Work

2.1 Older Adults and Information Threats Online

Research on older adults and their interactions with digital environments has been extensive, focusing primarily on how they navigate and utilize information related to security and privacy online [49, 56, 57]. Despite being increasingly active online, older adults often prefer traditional media and personal social networks over digital sources for obtaining cybersecurity information [57]. They tend to trust broadcast media and family members more than digital resources created by security experts [49]. Collaborative family efforts, such as relying on a "self-appointed family tech-manager," play a significant role in managing their digital security, affecting their motivation to independently engage with new technologies [56, 86].

Social engineering attacks, such as phishing and scams, are significant areas of concern for older adults [15, 17, 18, 29]. Studies have shown that older adults are aware of various social engineering tactics, particularly those exploiting their reliance on healthcare services [26]. However, despite this awareness, they often lack the technical support needed to effectively detect phishing attempts [60]. Older adults tend to use fewer phishing detection cues and may not recognize spam emails as unsafe [71, 85]. Nevertheless, their proactive approach to protect themselves from social engineering as a form of deception demonstrates a significant effort to maintain digital security. The challenges faced by older adults in digital environments extend beyond social engineering attacks [65]. Usability barriers, such as small text sizes and complex navigation, can hinder their ability to effectively engage with online content [59, 76]. These barriers highlight the importance of designing user-friendly interfaces that cater to the needs of older adults [16], enabling them to navigate digital spaces more confidently. This is especially important when it comes to manipulative narratives as social media platforms tend to overlook usability barriers when moderating misinformation [34].

2.2 The Threat of Online Misinformation

Misinformation has a particularly attractive pull around developing events where people turn to social media to seek information in real time [36]. Elections are such an event that develops during the campaigning period, the election day, and the aftermath, making it a fertile ground for developing and spreading manipulative narratives about the candidates [80]. English-speaking voters are the main target, though marginalized and immigrant diasporic communities are not spared either [53]. The goal of spreading manipulative narratives is to influence voting intentions by exploiting users' homophilic proclivity, content overload by interest groups on social media, and platforms' failure to moderate falsehoods [46]. That these manipulative narratives in the case of the

2015 UK Brexit and 2016 US Elections were actually able to influence the voting outcomes is perhaps hard to prove, but nonetheless, that doesn't diminish their threat to future elections [48]. For one, evidence shows that the effort put in their sustained dissemination in elections has increased since the aforementioned watershed events [47]. Taking this evidence into consideration with the fact that social media users are susceptible to misinformation [43], makes a compelling case to study how these manipulative narratives are experienced, internalized, and acted upon by the part of the social media population that is 65 years or older. This is especially important because data shows that older social users are more likely to be exposed to, click on, and re-share potentially misleading narratives than other user groups on social media [4].

2.3 Older Adults and Misinformation

The threat of misinformation on social media has inspired a focused research agenda examining how older adults engage with fabricated, deceptive, or sensationalist news headlines. Despite their increasing digital presence, older adults are frequently exposed to misinformation, particularly during significant events such as elections [30]. Evidence suggests that older adults can improve their ability to discern true from false news through digital literacy training and the use of fact-checking services [21, 52, 74]. This cautious approach to information consumption involves verifying news stories before accepting them as accurate [52]. Older adults often rely on familiarity and source heuristics when evaluating news [5, 83]. Despite the cognitive effort required, they frequently use fact-checking services, which helps them manage the vast amount of information encountered online. Although older adults tend to share and interact with fake news more than younger individuals [30, 31, 93], their accumulated life experience and knowledge enable them to better distinguish between true and false information over time [63]. They often stick with familiar knowledge and reject contradictory claims [6].

The social context of information is particularly significant for older adults, who place greater emphasis on cues about a person's character compared to younger counterparts [5]. This contextual evaluation is crucial in their process of discerning misinformation. Understanding how older adults interact with misinformation on social media is essential for developing targeted interventions and support systems. Enhancing digital literacy and resilience against misinformation can empower older adults to navigate digital content more effectively [51]. This research aligns with and extends the existing body of work by providing a deeper understanding of older adults' experiences with misinformation. It offers practical implications for improving their interaction with digital environments and supports the development of tools and strategies to help older adults become more discerning consumers of online information. Motivated reasoning, the tendency to process information in a biased way to protect one's political identity, affects individuals across all age groups, as evidenced during the COVID-19 pandemic [69], including older adults [72]. However, the ability to analyze potential information hazards improves with age, which helps older adults offset the challenges posed by misinformation [35]. This ability, combined with their life experience, allows older adults to develop more sophisticated strategies for identifying and mitigating misinformation.

2.4 Impact of Misinformation on Voting Intentions

Previous studies have extensively examined the impact of misinformation on political beliefs and voting behaviors [41, 54, 82]. However, the specific demographic of older adults has received less attention, despite the fact that 76% of voters aged 65 and above turned out to vote in the 2020 presidential election [10]. Misinformation could influence voting intentions by shaping individuals' perceptions and beliefs about political candidates and issues [63, 75, 87]. Older adults, as active participants in the democratic process, are particularly salient targets of misinformation that aims to influence their voting intentions. Studies have shown that exposure to false information can alter

perceptions and reinforce existing biases about political candidates, and with that, the perception of their fit and ability to hold office [45, 63, 92].

Older adults tend to share and engage with misinformation more frequently than younger individuals, underscoring the need to understand their specific vulnerabilities relative to shaping their voting intentions [31, 93]. Our study provides a comprehensive analysis of the factors older adults consider when encountering manipulative narratives on social media as the main battleground of information operations, including the perceived purpose of misinformation, cues of deception, reasoning strategies, and the broader social context. By highlighting older adults' opinions on the impact of misinformation on American society and voting intentions, our research offers valuable insights into the long-term effects of misinformation. This understanding informs the design of interventions to address these challenges. Enhancing digital literacy and resilience against misinformation can empower older adults to make informed voting decisions, thereby strengthening democratic processes and public trust.

3 Method

3.1 Research Questions

Understanding of how older adults experience and engage with misinformation on social media towards informing their voting intentions hasn't received much research attention. To this goal, we sought to answer the following five research questions.

- (1) **RQ1:** How older social media users with various voting intentions in the forthcoming 2024 US presidential election (*Democrat, Republican, Undecided, No Intention*) conceptualize misinformation in general?
- (2) **RQ2:** How older social media users with various voting intentions in the forthcoming 2024 US presidential election (*Democrat, Republican, Undecided, No Intention*) conceptualize the purpose of misinformation?
- (3) **RQ3:** How older social media users with various voting intentions in the forthcoming 2024 US presidential election (*Democrat, Republican, Undecided, No Intention*) discern truth from misinformation on social media?
- (4) **RQ4:** How older social media users with various voting intentions in the forthcoming 2024 US presidential election (*Democrat, Republican, Undecided, No Intention*) see misinformation affecting American society in general?
- (5) **RQ5:** How older social media users with various voting intentions in the forthcoming 2024 US presidential election (*Democrat, Republican, Undecided, No Intention*) envision misinformation evolving in the future?

3.2 Recruitment and Sampling

As shown in Figure 1, we first obtained approval from our Institutional Review Board (IRB) to conduct an exploratory survey and follow-up semi-structured interviews. The survey asked the participants to define misinformation in their own words (RQ1), elaborate on what they believe is the purpose of misinformation on social media (RQ2), detail their truth discernment strategies (RQ3), and provide their perspectives relative to how misinformation affects the American society in general (RQ3) and how they envision misinformation evolving in the future (RQ4). The interview expanded the survey to add rich accounts relative to the general experiences with misinformation and its influencing effect relative to shaping voting intentions.

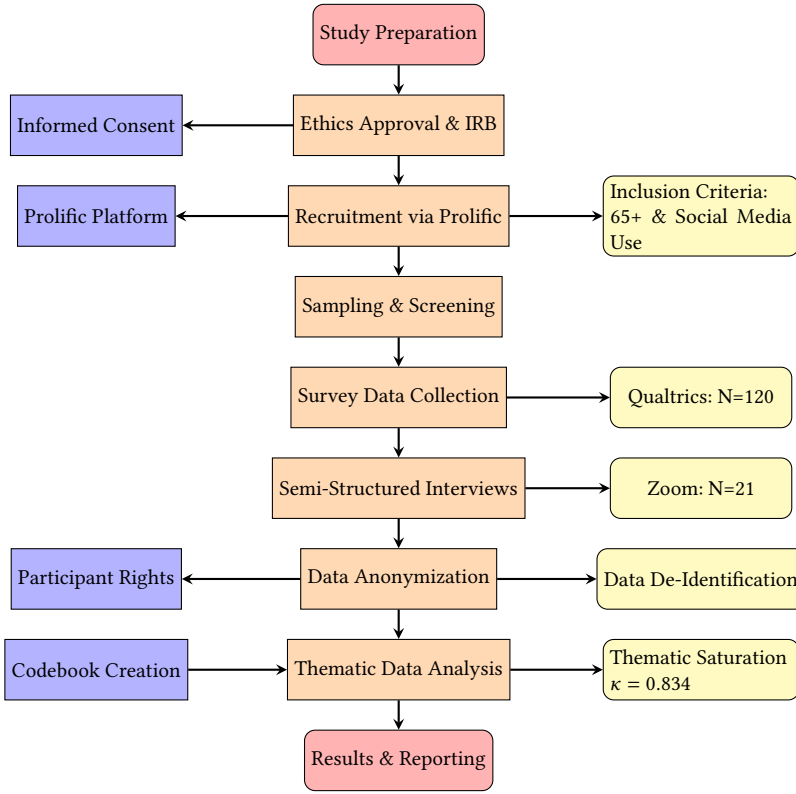


Fig. 1. Overview of the Research Methodology for the Study.

Our sampling criteria sought participants who are social media users ages 65 and above in the United States. The criterion “social media user” stipulated participants have at least one social media account and visit at least one social media platform regularly (daily or weekly) over a period of at least a year, reading at least a few posts per visit (there were no restrictions regarding posting, commenting, liking, or reporting posts on platforms as part of “usage”). We used Prolific for the recruitment, Qualtrics for the survey, and Zoom for the follow-up interviews. At the end of the survey, we offered each of the participants to take part in a flow-up semi-structured interview.

As our aim was an exploratory study, instead of a statistical test of any behavior around misinformation, we determined that a representative and feasible sample size of the older adult population in the US that participates on social media [23, 89], with a 5% margin of error and 95% confidence level was 120 participants. We invited 150 older adults and ended up with 119 participants after filtering out low-quality responses that didn’t answer any of the questions or provided immaterial answers. Out of these 119 participants, 21 agreed to do the follow-up semi-structured interview. The interview sample of 21 participants proved sufficient as we reached a thematic saturation, i.e., we collected data up to the point where there were fewer surprises in the responses to the research questions and no more emergent thematic patterns. Arguably, one could object that both the RQ4 and RQ5 allow for a practically unlimited number of perspectives relative to misinformation effects and evolution [61]. We acknowledge this issue in the limitations section further on, however, upon reflection, the perspectives converged to salient themes that we unpacked and presented with supporting quotes in the results section.

The survey responses were anonymous, but the interviews initially weren't. We offered the participants to use only the audio option in Zoom if they wanted and we recorded the meetings so we would get the automatic transcription ready for our later data analysis. Once we had the audio transcribed, we removed any personally identifiable information and checked the answers to establish full anonymity. Both the survey and the follow-up interviews allowed users to skip any question they were uncomfortable answering. The survey took around 20 minutes to complete and the follow-up interviews around 30 minutes. Participants were offered a compensation rate of \$4 for the survey and \$10 for the follow-up interview participation. The demographic structure of our survey and interview participants are given in Table 1 and Table 2, respectively. We achieved a balanced and diverse sample, leaning towards older adults with college or graduate degrees. The breakdown of social media platforms used by our participants (from both the survey and the interview) was: YouTube (82%), X (Twitter) (41%) Reddit (35%), Instagram (29%), LinkedIn (12%), TikTok (12%), Nextdoor (12%), and WhatsApp (6%). A further breakdown of the sample per age is provided in the [Appendix](#).

Table 1. Survey Demographic Distribution

Gender Distribution	
Female	64 (54%)
Male	54 (45%)
Non-cisgender	1 (1%)
Intent to Vote	
Democrat	68 (57%)
Republican	29 (25%)
Undecided	17 (14%)
No Intention to Vote	5 (4%)
Race/Ethnicity	
White	96 (81%)
Black/African American	17 (14%)
Identify with More than One Race/Ethnicity	3 (2%)
Latinx	2 (2%)
Asian	1 (1%)
Level of Education	
College	74 (62%)
Graduate	33 (28%)
High school	12 (10%)

Table 2. Interview Demographic Distribution

Gender	
Female	11 (52%)
Male	9 (43%)
Non-cisgender	1 (5%)
Intent to Vote	
Democrat	12 (57%)
Republican	8 (38%)
Undecided	1 (5%)
Race/Ethnicity	
White	17 (80%)
Black/African American	2 (10%)
Latinx	1 (5%)
Identify with More than One Race/Ethnicity	1 (5%)
Level of Education	
College	13 (62%)
Graduate	8 (38%)

3.3 Ethical Involvement, Risks, Safeguards

Aware of the sensitive nature of our study, it was important for us to establish trust with the participants, provide assurances about the protections we had in place, and offer support throughout the entire process of participation. We designed the study to be inclusive in that it considered the basic capabilities of older adults, their accessibility needs, it was relatable to them, and conducted in a way that was non-judgmental and empathetic [13]. As part of the process of obtaining consent in both the survey and the interview, we communicated that we to gather new insights and knowledge in a poorly understood and complex area such as lived experience with misinformation for older adults. We transparently noted that we won’t ask the participants to engage with any misinformation as part of the study, only to answer information about their past engagements and lived experiences. We deliberately avoided exposure to misinformation content to limit excessive distress or potential triggering effects directly as a result of the misinformative content.

We communicated with our participants that our agenda for change is focused on inclusive and safe social media engagement with misinformation for vulnerable populations. We acknowledged our own identity as researchers and the power within it as we are not older adults. We offered emotional and accessibility support on an ongoing basis both during the survey and the interview because talking about misinformation might be distressing and older adults might need accessibility accommodations. We thus allowed the participants to take breaks if needed (recording stopped) and then continue if they wished. Before and during the interviews, at all times, we gauged whether or not a potential participant was too vulnerable to take part in the first place. We did not have

such an instance, but our plan was to offer them an alternative time and date for the interview, in addition to any of the aforementioned meanings of support.

The informed consent process in the survey made it clear that the participation is anonymous, i.e., no names, addresses, social media accounts/handles, or any personally identifiable information is collected. Both for the survey and the interview, participants were informed that they were free to stop and abandon any question at any point in time and ask to stop the recording if they felt like doing so. Participants were notified at the end that the research team was available up to 30 days after they participated to attempt to locate and remove their data – as the participation was anonymous – if they were able to recall a unique way to identify their answers (none of the participants requested removal). Though anonymity was strictly safeguarded, we offered the option for the participants to share a contact of identifiable information to break confidentiality, especially during the interview, if they or we believed there to be a risk of harm, a need for immediate assistance, or an emergency (no such instance occurred).

It was important for us to ensure that we did not leave participants feeling worse after than they felt beforehand taking the survey and doing the interview. We extensively debriefed them that we don't act on behalf of, or receive any funding from, any social media platform. We informed that we are impartial relative to the use of warnings and that the determination of potentially misleading content is solely done by the platforms themselves. We, however, pointed participants to fact-checking such as FactCheck.com [22] or Snopes [79] where they could independently verify any claim or misinformation topic of their choice of their own volition. We also indicated that there is a possibility that after the publication of the results of the study, one could misuse or misinterpret them to reinforce stereotypes and negative discourses of age and aging relative to social media use and misinformation [5, 55]. We had no available safeguard against this risk, and we offered participants to withdraw their participation if they wished (no one did).

3.4 Data Collection

Previous studies have extensively analyzed the interplay between misinformation and people's political beliefs or attitudes [77]. Recently, this academic effort has come under increased political scrutiny, with claims and threats of lawsuits that the research on misinformation is driven toward suppressing conservative opinions [88]. To avoid the risk of further stereotyping relative to people's self-reported political stances, we chose to collect and group our participants by their *intention to vote* in the forthcoming 2024 US presidential election cycle. This way, we can allow for a better analysis of the political contextualization of misinformation without risks of misinterpretation as well as allow for situations where people who identify with one political ideology intend to vote for another party, remain undecided, or abstain.

Much of the prior research has also focused on older adults within specific regions by contacting older adults through senior centers [26, 73]. While this is an efficient method to gather data on older adults, we felt that a more widespread study would provide additional information on older adults who have at least some forms of digital literacy and are more likely to be habitual social media users. In addition, we felt it was important to include a larger, more diverse population of older adults because this aspect is often overlooked in other studies focused on older adults [32]. Capturing a diverse part of the population allowed us to provide an ecological validity to our qualitative findings relative to the older adults' misinformation mental models (concept and purpose), truth discernment strategies, and misinformation outlook.

3.5 Data Analysis

Our data analysis process involved several steps. First, we collated the interview responses with the responses from the survey, assigned numbers to the participants, and made a final check for

the anonymity of all answers. This ensured that no personally identifiable information was present in the dataset, maintaining participant confidentiality. We began with a round of open coding on the responses from two randomly selected participants. Open coding involved a line-by-line examination of the text to identify and label concepts and themes directly from the data. This initial coding phase was critical for developing a preliminary understanding of the participants' experiences and perspectives. Each researcher independently coded the responses to avoid bias and capture diverse interpretations.

Following the open coding, we held a series of internal discussions to compare and refine our coding schemes. We aimed to achieve consensus on the interpretation of the data, which led to the creation of an agreed **codebook** (listed in the Appendix). This codebook was structured to capture four main aspects: (i) *conceptualization* – codes pertaining to how older users mentally model misinformation in general; (ii) *purpose* – codes related to the purposes of misinformation that older adults believe are behind its spread on social media; (iii) *truth discernment* – codes describing the participants' cognitive strategies for discerning truth in the information they encounter on social media; and (iv) *misinformation effects* – what older adults believe are the effects of misinformation on American society in general. Next, we used the finalized codebook to conduct a comprehensive coding of the entire dataset.

Once the coding was complete, we moved on to thematic analysis. This involved identifying patterns and relationships between the codes to develop broader themes. We followed the thematic analysis approach outlined in [7, 8] and in [14], allowing us to construct a rich, nuanced narrative that connects the themes back to our research questions through data interpretation. For instance, in the *conceptualization* aspect, we identified themes related to the perception of misinformation as either outright falsehoods or as partial truths. In the *purpose* aspect, we found that participants attributed various motives to misinformation, such as political manipulation or financial gain. The *truth discernment* aspect revealed strategies such as cross-referencing multiple news sources and using fact-checking websites. Lastly, the *misinformation effects* aspect highlighted concerns about the erosion of trust in institutions and the potential for social division. Throughout the results, we will report on representative quotations to illustrate each theme. These quotations provide examples of the participants' experiences and perspectives, grounding our results reporting in the way we interpreted the responses conveyed to us.

4 Results

4.1 RQ1: Misinformation Mental Models

Past research shows that people's mental models of misinformation differ relative to whether they see misinformation as mainly falsehoods and inaccuracies that could easily be verified or fact-checked or a more nuanced form of deception that might contain elements of true and accurate information [77]. This distinction is important because in the latter case, the deception could be seen as a deliberate attempt to exercise influencing power over how people form opinions or decide about actions [11]. For this research question, we stopped short of explicitly asking about the intent behind the deception as we wanted to get more detailed and rich responses to the second research question. Therefore, here we only captured the way our participants conceptualized misinformation relative to a) *falsehoods and inaccuracies* (e.g., entirely lacking truthfulness); and b) *misleading or deceiving claims* (e.g., including elements of truth) [39].

We broke down the coded answers from both our survey and our follow-up interviews controlling for the participants' intention to vote in Figure 2. The majority of the participants that intended

to vote Democrat, or 37 (62% of misleading or deceiving claims group), think that misinformation contains elements of truth. For example, participant **P111** offered the following concept of misinformation during the follow-up interview:

“Misinformation is ‘information’ that is either entirely or partially untrue. It can be presented in a manner that makes it difficult to discern its veracity. Misinformation can be deliberately shared to convert people to a particular way of thinking, or it can be information taken out of context. Most often bias is involved. In the past, reputable publishers conducted fact-checking to limit the amount of misinformation shared, but the demise of traditional journalism and the spread of social media gives everyone a soapbox. People lacking the credentials to self-edit or those who deliberately seek to put out false info to promote their own agendas are quite prevalent.”

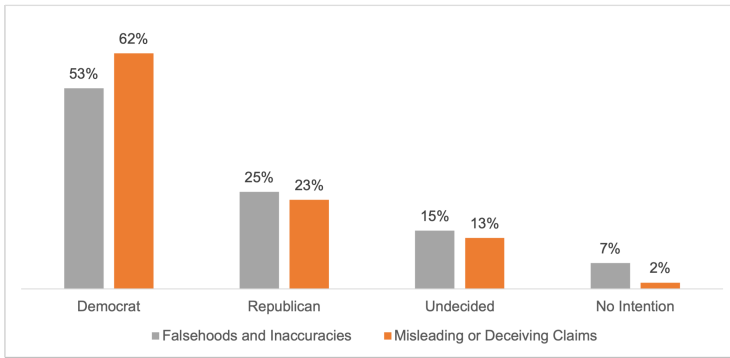


Fig. 2. RQ1: Mental Models vs. Intention to Vote

For most of the participants with voting intentions other than Democrat, misinformation was mainly falsehoods or inaccuracies. For example, participant **P40** simply stated that for them, misinformation is “*any information that has been verified not to be true.*” It is worth mentioning that we noticed a trend among 14 (23% of misleading or deceiving claims group) participants who intended to vote Republican to speak about “*disinformation.*” Here, **P35** offered this detailed account of disinformation:

“The word disinformation brings to my mind a government or reporting organization sponsoring or composing facts. Because I can remember back in the Cold War we, being the Western countries, would come out with probably disinformation about what was happening in Russia. And certainly, from what I can gather, the Russian media was the same in the other direction. They’re very happy to use, I hesitate to use the word propaganda, but I think that propaganda and disinformation intersect in that Venn diagram of reporting. To me, disinformation is a deliberate act by somebody. It takes a lot of planning to fabricate information and there is an agenda of some sort behind it.”

4.2 RQ2: Purpose of Misinformation

While in the first research question we were interested to learn whether older adults see elements of truth or not in misinformation on social media, in the second research question we wanted our participants to explicitly unpack the purpose of misinformation. Whether content is disseminated deliberately or not, studies differentiate between *misinformation* (no intentionality) and *disinformation* (intentionality). But determining intentionality is often difficult [84], so we thus considered

a more encompassing aspect of misinformation, i.e. its *purpose* on social media. This gave us an opportunity to ask our participants to elaborate more not just about whether there is intentionality or not, but who actually creates, disseminates, and benefits from misinformation in their opinions.

Past research offers an abundance of evidence to demonstrate that the purpose of misinformation is to create divisions and dissent along political lines [81, 96], perpetuate political counter-argumentation [77], and ultimately undermine political candidates for office [19]. But misinformation was also used and continues to be used outside of the political arena such as in health misinformation [28, 91], consumer product misinformation [97], and alternative narratives in general (e.g., conspiracies, rumors) [12, 27]. Equally, misinformation is an essential ingredient for a general propagandistic rhetoric of emotionally manipulative language [37]. Therefore, we analyzed the purpose of misinformation respective to a) *creation of divisions and/or dissent* (i.e., politically-related); b) *influence opinion and/or action* (e.g., mainly for personal or financial gain); and c) *manipulate emotions* (i.e., mainly for manipulating emotions).

As shown in Figure 3, regardless of the intent to vote, the majority of each group mostly saw that misinformation served the purpose of influencing opinions and actions. For example, **P100**, who intended to vote Democrat, acknowledged the political aspect of misinformation but offered a broader view on influencing opinions and/or actions:

“Yes, I think the two primary purposes are 1) to profit financially from misinformation that encourages others to purchase products, and 2) to “educate” others about issues the spreader thinks are important. This would include those who want to swing others to adopting specific views (e.g., on vaccines or abortion), voting for specific candidates, etc.”

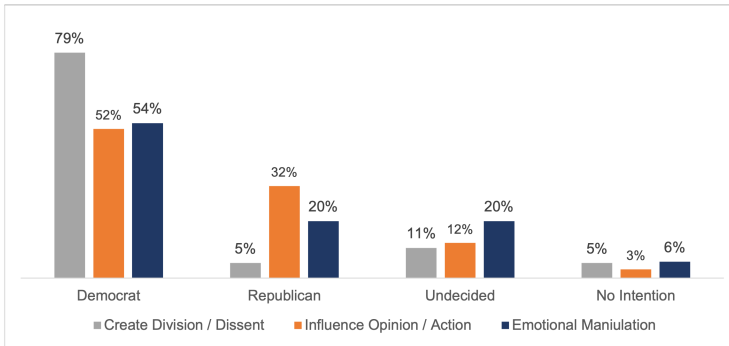


Fig. 3. **RQ2: Purpose of Misinformation vs. Intention to Vote**

However, we noticed that the participants with the intention to vote Republican in most of the cases concentrated on influencing opinions and actions or manipulating emotions as the main purpose of misinformation compared to only one (5% create division/dissent group) who believed misinformation has mostly a divisive/dissent purpose. For example, **P47** offered this take on the intent behind the spread of misinformation on social media:

“Of course the spread is purposeful. People distribute misinformation to hopefully change your mind about something or they want to gain power, money or whatever it is they desire to change. If you think social media is bad, try reading or watching any news articles that are totally biased.”

We also noticed that the 15 participants with the intention to vote Democrat form the vast majority of those participants who believe misinformation’s main purpose is creating political

divisions and/or dissent (79% within group). Participant **P63** believed that *“political groups intend to divide people by spreading misinformation towards influencing election results, fomenting racism, and inciting violence.”* Another participant, **P36**, stated that the purpose of misinformation is *“to sway popular opinion, to cause dissent, to widen rifts, to gain votes, to stir up trouble.* One of the participants we interviewed, **P10**, offered a bit more of a detailed take:

“Well, in the case of Russia, it’s been documented that they had not just a casual intent to spread misinformation on social media. I mean, there was a real campaign to get Donald Trump elected in 2016, and you know, probably also in 2020. But I know in 2016 it was definitely an orchestrated campaign. Yeah, I think that there’s always intent behind misinformation. I mean, they’re not just doing it for kicks. I think that you know they’re trying to mislead someone about an outlandish political candidate.”

Participant **P82** traced back the antecedents of the circumstances in which misinformation materializes its divisive and dissenting purpose:

“A lot of misinformation is designed by the corporations, the politicians, the billionaires to really just distract us and have us fighting among ourselves so they can just go and do whatever they want, make more money, control policy. And behind this is the intention of dumbing down us Americans. It started back in the sixties when the Nixon Administration purposely started underfunding public education. The kids that were raised in the sixties in the public school system, you know how old are they now? They’re my age. And we are the last generation that knows what critical thinking is. After us, the public schools taught an entire population to question nothing. And that’s why misinformation can get to these people because they’re like, well, that sounds good. No, it doesn’t. No, it doesn’t.”

Relative to the emotional manipulation purpose of misinformation, we noticed that the 7 participants who were undecided were prevalent here (20% within group), despite their smaller share in the overall sample. One of them, **P17**, stated that misinformation is *“designed to gravitate towards higher emotions and become more adamant about feeling that person’s theories about everything are always valid regardless if they actually are or aren’t.”* Those with no intention to vote were more leaning towards the emotional manipulation purpose with participant **P57** noting that *“in the US the purpose is to mislead and confuse, to leave the populace in ignorance so they can be exploited and made complacent”*.

We further queried and interviewed them to ask about who they think creates and disseminates misinformation and who benefits from it. The participants who were uncertain or who had no intention to vote pointed mainly to individual social media *“influencers”* as they either *“want credit for posting it,”* according to participant **P2** or *“gain notoriety from the shock “value” and the sheer malicious enjoyment of it,”* according to participant **P37**. The participants who intended to vote Democrat mainly blamed the *“other side”* of the political divide, or *“Trump and his supporters, white supremacists, Fox, right-wing media (Bannon, Jones), anti-gun control, Republican governors (Texas, Florida, etc.).”* according to participant **P18**. Participant **P26**, interestingly, offered a bit more balanced and extended answer:

“I see it most among conservative politicians and their followers. Trump followers are habitual misinformation hawkers but there are liberals on my side of the aisle that practice it too. QAnon followers are another group that heavily traffics in misinformation”

We noticed that the participants who intended to vote Republican were keen to blame the social media companies as the creators and distributors of misinformation. Participant **P108** pointed out *“the ‘higher ups’ in social media companies because they refuse to censor or correct information while*

deliberately permitting misinformation to be disseminated to attack their opponents". Participant **P24** extended this viewpoint as:

"The majority has been done by left-leaning organizations trying to keep the truth from different events and situations from coming out and of course they are helped by the algorithms and people who work hard to censor those who have a different viewpoint. The exception is now X."

As to who benefits from the misinformation on social media, those who intended to vote Democrat mostly pointed to "*big business and the GOP*", as participant **P15** put it. Participant **P38** extended this list to "*foreign governments trying to weaken the US; Political demagogic political operatives; Corporate interests like Exxon and Phillip Morris who flood the media with false information about the grave dangers of fossil fuels on climate or tobacco on public health, respectively.*" Those participants that intended to vote Republican, were uncertain, or planned not to vote saw the "government" as the main beneficiary of misinformation in addition to "*big tech and big pharma,*" according to participant **P101**. In the words of participant **P44**, the "*governmental agencies and the corporate elite benefit from misinformation because it keeps the public in the dark or they provide wrong data to get the public to do what they want.*"

4.3 RQ3: Truth Discernment

With our third research question, we wanted to learn how older adults discern between misinformation and truthful statements on social media. Past research suggests that poor truth discernment is linked to a lack of analytical reasoning and relevant knowledge, as well as to the use of familiarity and source heuristics [64]. At some point, it was believed that people are "better" at discerning truth from falsehood (despite greater overall belief) when evaluating politically concordant news [42, 90]. However, more and more evidence showed that "better" truth discernment is associated with analytical reasoning, regardless of whether a statement is consistent or inconsistent with one's partisanship [3, 9]. To this end, we probed our participants about the cues and the reasoning they employ when assessing content on social media or whether they mainly employed a) *heuristics* (e.g., familiarity or source); b) *analytical thinking* (e.g., they reflected or deliberated about the content); and c) *a mix of both strategies*.

The main theme that emerged from both our survey and the interviews was that regardless of the intention to vote – shown in Figure 4 – our participants mainly relied on the source heuristic in combination with fact-checking to discern truth from misinformation. Interestingly, many participants mentioned they use additional sources like the BBC to also check the reporting on US politics because, as participant **P39** put it, "*they are less prone to American bias*". The main rule of thumb, perhaps, was the use of familiarity with the topic and the source heuristics to look for cues of fabrication or deception. Participant **P26** who intended to vote Democrat explicitly referred to the source heuristics:

"I look at the source of the information in terms of who is stating the information. I know if it originates from a political source not to take it at face value. Same for partisan influencers. Even comments from private individuals make me look back at their other comments to see what they've said in the recent past. Doesn't take long to see who is ultra-partisan and who is not."

Another participant from this group, **P100**, elaborated on their combined tactics for discernment between misinformation and truthful statements based on their familiarity with emotionally manipulative language:

"I look for emotion-based, sweeping statements that offer no factual evidence or that offer evidence I don't see as credible. Especially if they're trying to sell something or stir up

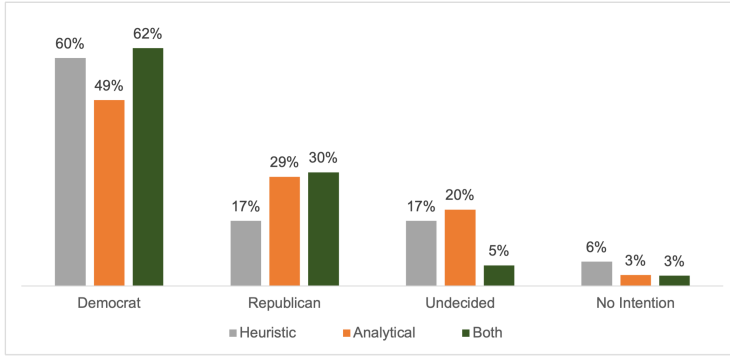


Fig. 4. RQ3: Truth Discernment vs. Intention to Vote

enmity toward certain groups, races, public health workers, public officials, etc. I'm alert for suggestions of violence against others or property. If I'm not sure about a claim, I check with sources I'm certain are credible, such as the CDC, state or local public health departments, evidence-based science journals, my city councilor, health care professionals, etc."

Participant **P62** explicitly mentioned the use of fact-checking services as part of their varied analytical approach in spotting any content that might be entirely lacking or omitting good parts of the truth:

"I try to use mainstream media such as the Washington Post, or the New York Times for information that I want to be factual. I also keep up with the local news and carefully choose which affiliate I am watching. It did not take me long to figure out that broadcast owners broadcast what is in their best interest and TV can be slanted to the liberal or conservative side. I recently signed up for a fact-checking newsletter to stay better informed about what is or is not the truth. I think it is very easy to become misinformed by social media simply because so many people with so many different points of view are participating with each one having their own agenda."

The participants who intended to vote Republican employed the same strategies for truth discernment as their Democratic counterparts. Participant **P35** offered an interesting approach using the source heuristics:

"Confirmation bias is always a problem here, where information that generally accords with one's own beliefs is likely to be accepted even if it includes glaring inaccuracies. I look to see if claims are patently ridiculous, i.e., Donald Trump eats babies. The spelling and grammar of a printed source are as important as sloppy work may conceal sloppy facts. What is the source? Are they generally reliable? Are they positioned to the left or right of the political spectrum? Finally, does it coincide with what I reasonably believe to be true?"

A noticeable portion of the participants who intended to vote Republican indicated they regularly turn to fact-checking services. Using their familiarity with emotionally manipulative language in combination with reflective reasoning, participant **P117** said that *"fact-checking sites are a great way to check the credibility of news I come across because stories that have a lot of emotional manipulation tend to be false."* Another participant, **P85** seconded this approach, adding:

“I look to see who posted the misinformation - do they post only about one thing or how many followers do they have? If something does not sound right, I will do a fact check on Snopes or Politifact”

The participants who were undecided or intended not to vote were also keen on using the source heuristics or turning to fact-checking services, many of them invoking the lack of critical thinking in sharing fake news on social media. Participant **P37** pointed out that they know a post is misinformation *“when it has a high level of stridency, it has absolutely nothing to do with critical or factual thinking, and people persistently post the same content (verbatim) from other “sources”*. Another participant, **P54**, amended this strategy, stating:

“I always look for credible sources, or I will use Politico or Snopes or others to vet it myself. I think it is vital nowadays to be very discerning when reading about something on social media. Look at who posts the info, is it from a credible source, and even if it is someone you trust, a quick search to vet it is almost necessary to do so in these times.”

4.4 RQ4: Misinformation and Society

With the fourth research question, we probed our participants on their opinions on how misinformation, or the general trend of truth decay, affects American society in general. The truth decay phenomenon is associated with a set of several specific trends, namely: (a) *motivated reasoning* or reasoning driven more by personal or moral values than objective evidence and analytical interpretations of data [42]; (b) *escalation of extremism* - or the collaborative construction and amplification of alleged evidence of general foul play (e.g., election fraud) used to facilitate extremist action [67]; and (c) *trust erosion* or diminishing trust in formerly respected institutions as sources of factual information [68]. Respective of these specific trends, we summarized the coded answers of our participants' combined answers in Figure 5.

As to the intention to vote, we noticed that 28 (74% within group) participants who intended to vote Democrat were of the opinion that misinformation will escalate extremist actions, but less concerned that it will exacerbate the trend of motivated reasoning ($n=27$, 50% within group) and further and erode the trust in formerly respected institutions ($n=13$, 48% within group). Participant **P19** expressed the following concerns:

“It’s dangerous to society. During the pandemic, it probably caused illness and more deaths by misleading people as to how they needed to protect themselves against COVID-19. In politics, it can cause the rise of extremist leaders and governments, convince people of false election results, suggest that people have done despicable things they didn’t, or cause people to think someone is innocent who is a criminal and dangerous.”

Another participant, **P36** expressed concerns about the erosion of trust in institutions in the context of widening the divisions among Americans:

“I think misinformation divides our society more and more, widening the gap in political differences. It also makes people not trust the government, the media, education, science, and more. It causes us to fight against each other, rather than join forces for the common good.”

Those participants who intended to vote Republican or were undecided were mostly concerned that misinformation would both exacerbate the trend of motivated reasoning as well as further erode the trust in the institutions. Participant **P35** offered the following opinion:

“I think society is much worse off because of disinformation. Before the Internet, if you wanted to fact-check something you went to the library and looked it up. Since more effort was involved in writing a book, and publishers valued their reputations, there was

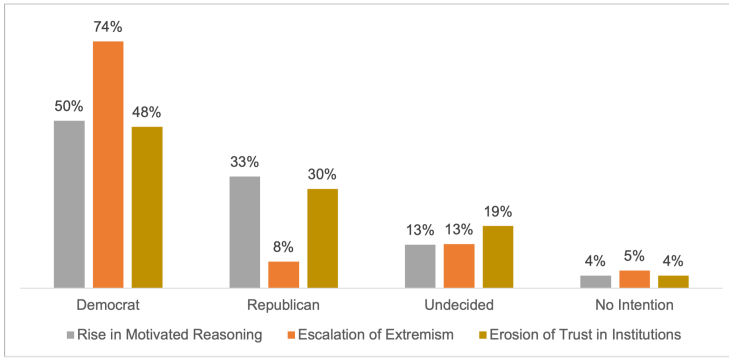


Fig. 5. RQ4: Misinformation and Society vs. Intention to Vote

an incentive to get things right. Now anyone can write anything and instantly have an audience of potentially millions. Some percentage of that audience will be stupid and blithely pass along any old trash that appeals to them. Society as a whole is composed of people from all walks of life and if a certain number of them believe things that are just wrong it is bound to have a negative effect on any collective decisions in the long run.”

4.5 RQ5: Misinformation in the Future

We came up with our final research question after we analyzed the results of the survey and identified that older adults as a target demographic could offer rich reflections on how misinformation might evolve in the future, based on their unique experience with misinformation – in any form – over a longer period of time and even preceding the Internet and social media. Therefore, we decided to ask those participants who opted for a follow-up interview to expand upon how misinformation might evolve relative to a) potential *public health crises in the future*; and b) *future election cycles* starting with the forthcoming one in 2024.

The main theme we noticed among our participants was skepticism that people in the US will re-establish their trust in the institutions, particularly the health authorities. As one of the demographics that suffered the most during the COVID-19 pandemic, our participants used their experience with the virus and vaccination misinformation [75, 84] to reckon about future public health misinformation. In these regards, participant P35 stated:

“I think that because of what happened during COVID people are much more skeptical now about what they are being told to do and not do. I guess more specifically if the government says, run down and get a vaccination or always wear a mask or never, ever, ever leave your house or try not to breathe. I believe there’s going to be a lot more skepticism involved, and I think if it’s another COVID which is going to be with us forever, everybody will catch it eventually. Everybody will build up some sort of immunity to it, but I think if it’s something far more serious, like Marburg Virus Disease or some of the viruses that are emerging in North Africa and are now popping up around the world, there won’t be the public response to effectively quarantine, vaccinate and defend against it because of the mistrust that a lot of people now feel.”

Participant P14, invoking the politicization of the COVID-19 pandemic [33] and the specific trend of rejection of science [68], pointed to the antecedents of the general health misinformation as a setup that will enable it to continue negatively affecting public health with a very honest account:

“[Misinformation] initially increased a lot of disbelief in science and will continue to do so. I think one of the ways I look at it is public health is not the strong side of the United States because you have so many institutions against each other. It was the CDC against the NIH, and this one here, and this one there. I am an immunologist and of course I believe in vaccinations, but I was skeptical when the government said it would take a year to develop a COVID-19 vaccine because previously it would take 5 or 6 years to develop one. Early on in the vaccine development and testing process, I thought all of it might well be misinformation, but the mistake was that neither the CDC nor the NIH were open to further simple explanations and transparency to let people know that it was a development and learning in progress and that the vaccine will anyhow evolve over a longer period time. So, these misinformation mistakes might happen again in the future and cause another disaster.”

Focused on the forthcoming 2024 election and evoking the past two election cycles, most of the participants could not avoid involving Donald Trump when they surmised about the election misinformation future. For these participants, he may have had (or will have) a temporary role in the misinformation conundrum but he has done permanent damage to overall political affairs in the US, especially with his brand of motivated reasoning. Participant **P23**, to this end, stated:

“I’m frightened. I think misinformation will very possibly incite violence. There’s not going to be any truth coming from anywhere, because the truth is just off the table. Now, nobody cares. People want to feel like they know what’s going on, but they don’t. I think it’s very frustrating for everyone. If someone tells you, they know what’s going on politically, automatically you know that you probably can discount whatever they say. It’s just not true. We live in a greed-based society that kind of drives everyone to use misinformation to get what they want. And it will get worse in every future election cycle.”

5 Discussion and Implications

Our findings indicate older adults are fairly well attuned to the current state of misinformation on social media. Although we didn’t explicitly test any particular misinformation content nor employ any truth discernment interventions, we found substantial evidence that older adults extensively research both news stories and other types of content before deciding if they were accurate or not. This suggests a proactive approach to handling misinformation, with a strong determination to use fact-checking sites and, interestingly, foreign journalistic services in combination with source and familiarity heuristics. Our sample, which included a majority of older adults with college or graduate degrees, showed a consistent pattern of independent analytical thinking. These individuals sought expert opinions *outside* of their social groups, rather than relying solely on familiar sources within their social circles. To these points, participant **P57** noted:

“Yes. I typically trust Snopes.com to be unbiased and fair. There are others, like Factcheck.com. Otherwise, I fact-check myself by visiting multiple well-known news sites, and try to read things from central, left-leaning, and right-leaning sources to keep a fair balance.”

We found evidence that older adults reject claims that contradict their knowledge, even when these falsehoods feel familiar. Additionally, older adults consider the social context, such as cues about a person’s character. During our follow-up interviews, five out of eight participants who intended to vote Republican stated they are fully aware that Republican or conservative voices often speak misinformation, even though they are closely aligned with their political ideology. Participant **P93** honestly remarked:

“I like Trump. I thought he did a good job, but you know he’s misleading people. He is telling them, ‘They are after you’. There’s nobody after you or me. So that’s misinformation. I

don't even think he's smart enough to even concoct that up. I think somebody wrote that for him and said, 'say this and your poll numbers will come up'. Okay, that's misinformation."

Our work highlights that older adults, in considerable numbers, model misinformation as a form of political (counter) argumentation that often contains elements of truth. The older adults in our sample saw the "other side" of the political divide as responsible for misinformation on social media, even though they acknowledged that there are spreaders *on their side of the aisle*, as participant **P26** said. The concerns about the more pronounced tie between extreme polarization and misinformation adherence were also expressed in our study. Altogether, the outlook offered by the participants in our study seems to be rather non-optimistic as many of them were skeptical of restoring the democratic order they remember from the past.

While the older adults in our study demonstrated a significant investment in discerning the truth from misinformation, this proactive approach may not be representative of the entire older adult population. Particularly, those who require assisted social media access might not have the same level of engagement or capability in discerning misinformation. This disparity highlights the need for social media platforms to adopt more effective strategies to assist these users in navigating misinformation. The nature of truth discernment for the older adults in our sample heavily relied on external sources outside of social media. This reliance indicates a gap in the platforms' ability to provide adequate tools and resources for users to independently verify the accuracy of information. This gap is critical, as the effectiveness of information dissemination on social media directly impacts users' ability to make informed decisions, whether concerning voting outcomes or public health issues. A significant concern voiced by participants in our study was the erosion of trust in institutions, including social media platforms. This erosion of trust stems from the platforms' perceived inability to address misinformation effectively. The lack of collaborative, independently verifiable, and accessible interventions exacerbates this issue, highlighting the need for a multi-faceted approach to misinformation management on social media.

5.1 Designing Collaborative Interventions

To address the cooperative deficiencies in social media participation among older adults, we propose the development and implementation of interventions that enable: (1) pre-sharing credibility assessment; (2) contextual truth discernment; and (3) an ability to participate in misinformation debunking. These interventions are designed to leverage the active engagement of older adults in credibility assessment and truth verification, as evidenced by our study. Pre-sharing credibility assessment interventions could involve interactive prompts that encourage users to reflect on the source and content of the information before sharing. This might include brief quizzes about the content's source, designed to make the user pause and consider the credibility of the information.

For instance, a prompt could ask, *"Is the source of this article reputable? Have you checked the facts?"*. The pre-sharing credibility assessment could be well aided by automated means such as CredEye tool, proposed by Popat et. al [66], where the source and content of a post could be verified against relevant articles from the web to deliver a credibility verdict about the included statements. This verdict could accompany each post as a label underneath each post, as suggested by Sharevski's et. al [78] in their design of an experimental social media platform for calibrating a misinformation-ridden discourse. Platforms could also implement a rating system where users can see the credibility scores given by other users before deciding to share content. This system would create a community-driven effort to assess the credibility of information, encouraging users to think critically about what they share.

5.2 Collaborative Pre-Sharing Credibility Assessment

We believe that the investment in credibility assessment could be harnessed towards *collaborative pre-sharing credibility assessment* where users are not just asked to check a box but informed how people on the platform (or alternatively, a selection of in/out communities) responded to the intervention before. These improved interventions bring together the benefits of the sharing timeline and the “community notes” where users with both left-leaning and right-leaning political worldviews work towards labeling potential misinformation. A study conducted by Drolsbach and Pröllochs [20] shows that the crowdsourced fact checking or the “community notes” are well received as an intervention, with many social media users agreeing with the community-generated fact-checks. A further exploration of the crowd-sourced fact-checking participation conducted by Saaed et al. [70] uncovers that the users usually reach agreement on highly misleading political posts and that the fact-checking is on par with the fact-checking done by experts.

Our results suggest that older adults are uniquely positioned and equipped to participate and lead the discernment and implicit labeling effort as suggested in [70]. Older adults, our study shows, are trustworthy users who see beyond the confirmatory aspects of narratives relative to the dominant political ideologies in the US, a threat to the crowdsourced fact-checking identified by Saaed et al. [70]. This form of *contextual truth discernment* in equal capacity serves in labeling posts that already have shared misinformation. In this case, the timeline is expanded with an explanation of why a post contains misleading narratives, developed through a consensus-driven, crowdsourced “fact checking.” Collaborative pre-sharing credibility assessment can be enhanced by creating user groups that specialize in different topics, such as politics, health, or science.

These groups could provide ratings and annotations on articles that fall within their expertise. For instance, a group of retired scientists could rate the credibility of scientific articles, while a group of former journalists could assess news stories. This specialization would add a layer of expert validation to the pre-sharing process, making it more robust and reliable. It will also address another threat identified by Allen et. al [2] where non-expert users often struggle to separate their political or ideological biases when evaluating claims, leading to disputes and biased crowdsourced fact-checks. Additionally, platforms could implement a peer review system where users can submit articles for review by their peers before sharing them. This system would not only encourage users to think critically about the information they share but also create a sense of community and collaboration among users. Along these lines, Xu et. al [94] and Jahanbakhsh et. al [38] found that fact-checkers are receptive to such reviews and external input.

5.3 Participating in Misinformation Debunking

The development of this explanation, or the debunking of the misinformation, could greatly benefit from the “wisdom” of the older adults, our results suggest. Because the actual truth could be more complicated, i.e., feeling less appealing than sensational misinformation, older adults as debunkers are ideally suited to transform complex ideas into easily imagined and readily comprehensible terms for wider audiences, resulting in explanations that could well sustain the threat of implicit political bias. The ability to *participate in misinformation debunking*, in our view, is also beneficial to older adults in that it creates a collaborative resilience against influences on voting outcomes by allowing this population to engage early – rather than later – with fact-checking of emerging historical, scientific, and health information operations narratives. The opportunity for an active participatory nature, considering the concerns expressed by our participants, could well aid a platform-driven (mis)information literacy that would counter the rise in motivated reasoning in American society.

Participating in misinformation debunking could involve training programs where older adults learn how to identify and counter misinformation. These programs could be offered by social

media platforms or independent organizations and include modules on critical thinking, media literacy, and fact-checking techniques. Additionally, creating online communities where older adults can collaborate and share their debunking efforts would foster a sense of purpose and community engagement, further empowering them to combat misinformation. Moreover, platforms could develop dedicated spaces for older adults to engage in debunking activities, providing them with the tools and resources needed to effectively counter misinformation. For example, Zade et al. [95] proposed a “Twitter trajectory” tracking tool that helps users visually identify and counter misinformation. We conjecture that older adults would be receptive to such tools, given that we found a significant investment in truth discernment in our study.

5.4 Enhancing Digital Literacy and Critical Thinking Skills

Our study highlights the need for targeted interventions to enhance digital literacy and critical thinking among older adults. Given their reliance on external sources for truth discernment, platforms can develop comprehensive digital literacy programs tailored to older adults. A comprehensive review of the essential elements to build these programs is offered by Aghajari et al. [1] and Hartwig et al. [34]. These programs, in essence, should cover essential skills such as identifying credible sources, understanding the indicators of misinformation, and using fact-checking tools effectively. Interactive workshops, online courses, and community-based training sessions could be organized to reach a broader audience. Additionally, integrating digital literacy content into existing community centers or senior organizations can provide a supportive environment for older adults to enhance their skills.

Moreover, gamified learning experiences could be developed to make the acquisition of digital literacy skills more engaging. One such option is the game “Fakey” developed by Micallef et al. [50]. The game effectively primes “players” to be suspicious of articles from questionable sources and we believe that older adults would find it useful and fun to play, given its simplicity and educational utility. In addition, creating interactive apps that simulate real-world scenarios involving misinformation can help older adults practice their skills in a controlled environment. These apps could include quizzes, puzzles, and role-playing games that reward users for correctly identifying misinformation and using fact-checking tools. Along these lines, Jeon et al. [40] proposed a controlled game aimed at increasing a player’s awareness of and preemptively responding to misinformation-prone echo chambers on social media. Additionally, partnerships with local libraries and educational institutions, as well as community involvement could facilitate access to digital literacy resources, ensuring that older adults have the opportunity to improve their misinformation-spotting skills. Kropczynski et al. [44] showed that communities of older adults work together to collectively manage their digital privacy and security, and we see this as an ideal opportunity to do so relative to misinformation.

5.5 Creating Trusted Networks and Information Gatekeepers

To address the erosion of trust in institutions, including social media platforms, our findings suggest the establishment of trusted networks and information gatekeepers within the older adult community. These networks can consist of respected community leaders, retired professionals, and subject matter experts who can act as reliable sources of information. For example, Nicholson et al. [58] proposed an initiative to support older adults to become community cybersecurity educators (CyberGuardians), tasked with promoting cybersecurity best practices within their communities to prevent older adults falling victim to opportunistic cyberattacks. We see this as an ideal basis to develop such an initiative for older adults as MisinformationGuardians. Platforms can facilitate the creation of these networks by providing the necessary infrastructure and tools for communication and collaboration and foster the creation of communities as identified by Kropczynski et al. [44]. For

example, dedicated forums or chat groups can be set up where older adults can discuss and verify information before sharing it widely. Additionally, appointing information gatekeepers within these networks who have undergone training in misinformation detection and fact-checking can further enhance the credibility of shared content.

Furthermore, these networks can be bolstered by regular virtual meetings and webinars where experts discuss current misinformation trends and provide guidance on how to navigate them. Platforms can also implement features that highlight content verified by these trusted networks, making it easier for users to identify reliable information. By fostering a sense of community and trust, these networks can serve as a valuable resource for older adults seeking reliable information and help mitigate the spread of misinformation. To ensure the effectiveness of these trusted networks, platforms could introduce a mentorship program where tech-savvy older adults or younger volunteers guide their peers through the complexities of digital literacy and misinformation detection. This mentorship approach would not only enhance the digital skills of older adults but also promote inter-generational solidarity and knowledge sharing, further strengthening the community's resilience against misinformation.

6 Future Work and Limitations

We intentionally focused our research on US social media users aged 65 and above to examine their dynamic political nature and varied voting intentions. However, our findings highlight the need to broaden our sampling to include older adults with diverse backgrounds. In future extensions, we plan to recruit participants with various levels of education, as our sample predominantly consisted of individuals with college or graduate degrees. We also plan to diversify our samples relative to household income, work experience, nuanced social media participation (e.g., interaction-based networks such as Twitter, Facebook; viewing-based networks such as TikTok and Instagram; and professional-based networks such as LinkedIn), and geographical area of residence. Additionally, we did not test self-reported mental models and truth discernment strategies with actual social media content as that requires a controlled lab setup and expedited research protocol approval, which we plan to do as a follow-up to this study. An important limitation also comes from the fact that we didn't measure the cognitive decline/acuity for our participants and thus cannot provide any details about its relation to our findings. We also refrained from adding additional analysis relative to other demographic factors to avoid possible misinterpretation of our data, especially about dimensions such as gender and race. We leave this analysis for our future work.

We plan, in the future, to also explore how changes in social media platforms and policies impact older adults' perception and discernment of misinformation, as well as the self-directed debunking mechanisms they use. We will expand our study to include diverse platforms like TikTok and consider the role of assistive technologies to provide a more comprehensive understanding of older adults' interactions with misinformation. Our goal is to go beyond the predominately text-based misinformation and explore the experiences older adults have with audio, video, and image deepfakes, Large Language Models (LLMs), and any new forms of synthetic multimedia used for manipulative purposes. We also aim to develop and test "misinformation literacy" interventions tailored to older adults to enhance their resilience against misinformation and support informed decision-making in democratic processes. Despite the limitations, our study provides valuable insights and a foundation for future research aimed at improving digital literacy and trust in online information.

7 Conclusion

Our study provides critical insights into how older adults with different voting intentions conceptualize, discern, and perceive the impact and future evolution of misinformation on social media in the

US. By conducting a survey with 119 participants and follow-up semi-structured interviews with 21 of them, we found significant variations in their mental models and strategies for dealing with misinformation. Older adults intending to vote Democrat were more likely to view misinformation as containing elements of truth and aimed at creating divisions, while Republicans often perceived it as falsehoods designed to influence opinions or manipulate emotions. Regardless of their voting intentions, participants employed a combination of heuristic and analytical strategies to discern truth, heavily relying on source credibility and fact-checking services like Snopes and trusted sources such as the BBC, which is a non-domestic public service broadcaster. The implications of our findings highlight the need for targeted interventions to enhance digital literacy and critical thinking among older adults. Specifically, our work underscores the potential in designing tools and platforms that support older adults in recognizing and mitigating the effects of misinformation. Developing educational resources that emphasize the importance of source credibility, creating user-friendly fact-checking tools that are easily accessible, and tailoring media literacy programs to address the unique challenges faced by older adults can significantly contribute to promoting a healthier information ecosystem and fostering a more informed society.

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APPENDIX

A Codebook

- **Conceptualization** Codes pertaining to how older users mentally model misinformation in general:
 - **Falsehoods and Inaccuracies** The participant defined misinformation as information that lacks truth and truthfulness [39]
 - **Misleading or Deceiving Statement** The participant defined misinformation as information that contains elements of truth to mislead or deceive [77]
- **Purpose** Codes pertaining to how older adults see the purposes of misinformation behind its spread on social media:
 - **Create Divisions / Dissent** The participant expressed that misinformation’s main purpose is to create divisions and cause political dissent [81, 96]
 - **Influence Opinions and Actions** The participant expressed that misinformation’s main purpose is to influence opinions and actions [91]
 - **Propagandize** The participant expressed that misinformation’s main purpose is to support propagandistic efforts [37]
- **Truth Discernment** Codes pertaining to how older adults discern truth from misinformation on social media
 - **Heuristics** The participant expressed that they mainly employ heuristics, e.g., cognitive shortcuts, intuition, and feelings [64]
 - **Analytical Thinking** The participant expressed that they mainly employ analytical thinking [3]
 - **Both** The participant expressed that they employ a mix of heuristics and analytical thinking
- **Misinformation Effects** Codes pertaining to the older adults’ perspectives on the effects of misinformation on American society in general
 - **Rise in Motivated Reasoning** The participant expressed that they see negative effects such as a rise in motivated reasoning across the entire population of the US [42]
 - **Escalation of Extremism** The participant expressed that they see negative effects such as escalation of extremism within the US [62]
 - **Erosion of Trust in Institutions** The participant expressed that they see negative effects such as continued erosion in trust of the governing institutions in the US [63]

B Age Breakdown

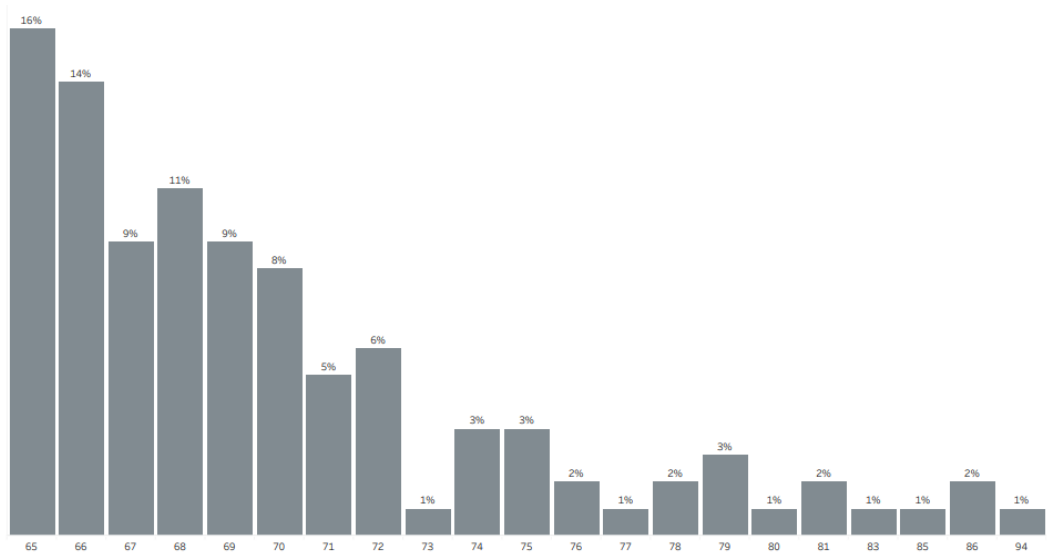


Fig. 6. Participants' Age Breakdown: Survey

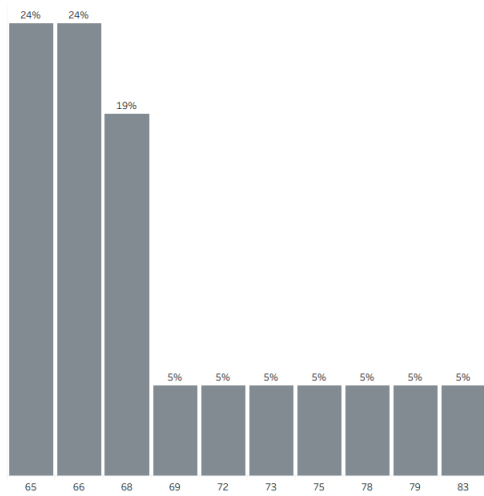


Fig. 7. Participants' Age Breakdown: Interview

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